Spectral Simulation Report

Automated Analysis

June 18, 2025

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1 Introduction

This report presents the results of spectral simulations derived from HDF5 dataset analysis. Each sample includes a spectral plot displayed in a full horizontal page and a detailed table summarizing the initial composition, actual positions, total pixels, and signal-to-noise ratio (SNR), along with simulation parameters such as temperature and electron density.

2 Sample: 12098

2.1 Simulation Parameters

Parameter	**Value**
Temperature (T)	10657 K
Electron Density (n_e)	4.53e+15 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	47.04

Table 1: Simulation parameters and additional metrics for sample 12098.

2.2 Composition Analysis

Element	**Initial Composition (%)**	**Actual Positions**
Al	17.44	462.0
Ar	0.00	0.0
C	0.00	0.0
Ca	17.25	150.0
Cl	15.49	35.0
Co	0.00	0.0
Cr	4.88	397.0
Fe	0.00	0.0
Mg	4.21	352.0
Mn	16.98	706.0
N	0.00	0.0
Na	2.39	322.0
Ni	6.40	478.0
O	0.00	0.0
S	11.11	228.0
Si	3.85	221.0
Ti	0.00	0.0
background	N/A	1847.0
Total Positions		5198.0

Table 2: Composition analysis for sample 12098.

2.3 Spectral Plot

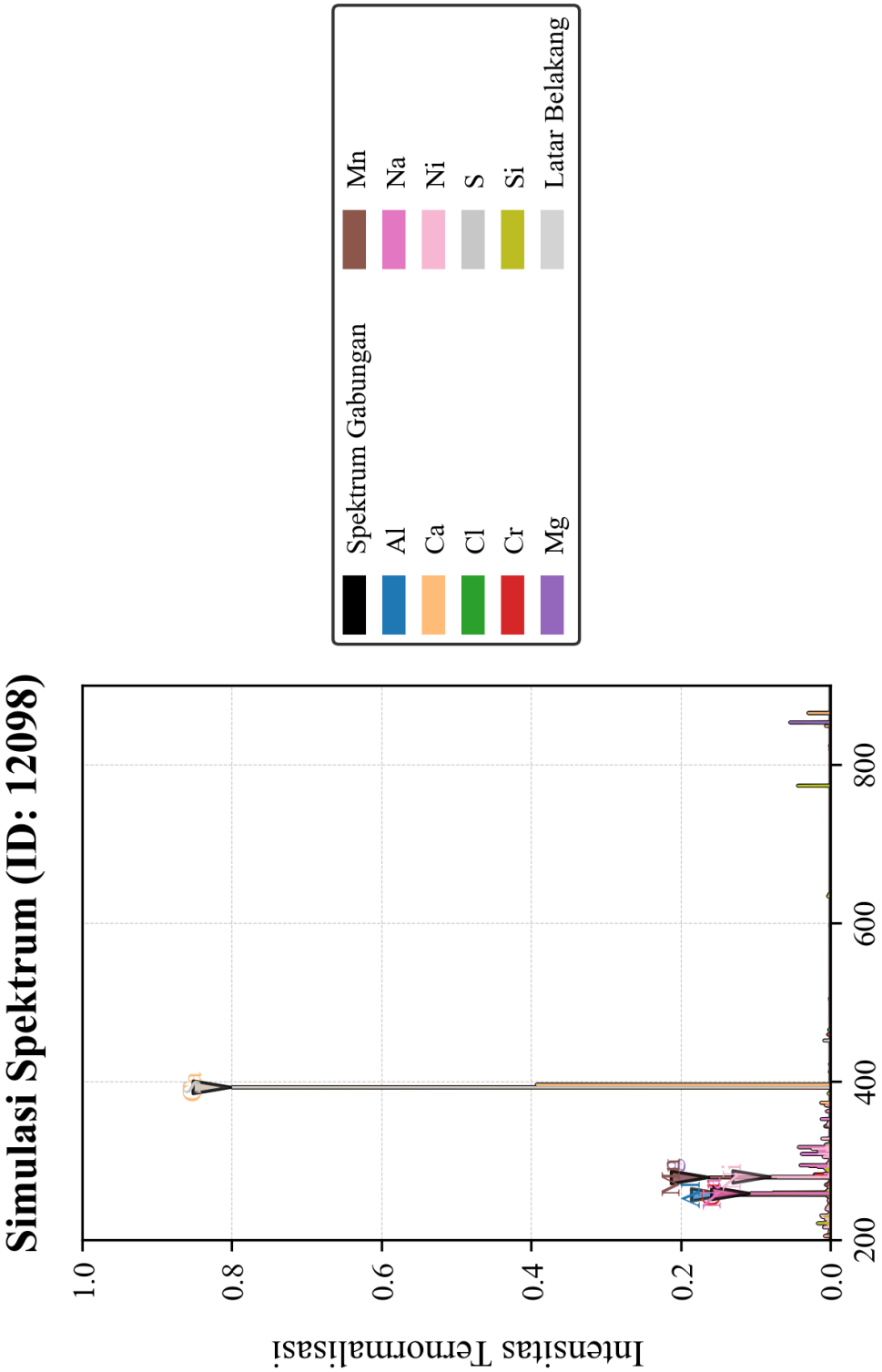

Figure 1: Spectral simulation plot for sample 12098 in full horizontal layout.

3 Sample: 15240

3.1 Simulation Parameters

Parameter	Value
Temperature (T)	7929 K
Electron Density (n_e)	4.98e+15 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	35.07

Table 3: Simulation parameters and additional metrics for sample 15240.

3.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	4.11	287.0
Ar	10.89	725.0
C	5.07	224.0
Ca	11.61	150.0
Cl	3.67	12.0
Co	0.00	0.0
Cr	10.33	397.0
Fe	0.00	0.0
Mg	4.64	326.0
Mn	2.97	706.0
N	2.81	396.0
Na	9.54	325.0
Ni	4.60	478.0
O	9.70	117.0
S	8.18	77.0
Si	5.14	218.0
Ti	6.75	1184.0
background	N/A	1144.0
Total Positions		6766.0

Table 4: Composition analysis for sample 15240.

3.3 Spectral Plot

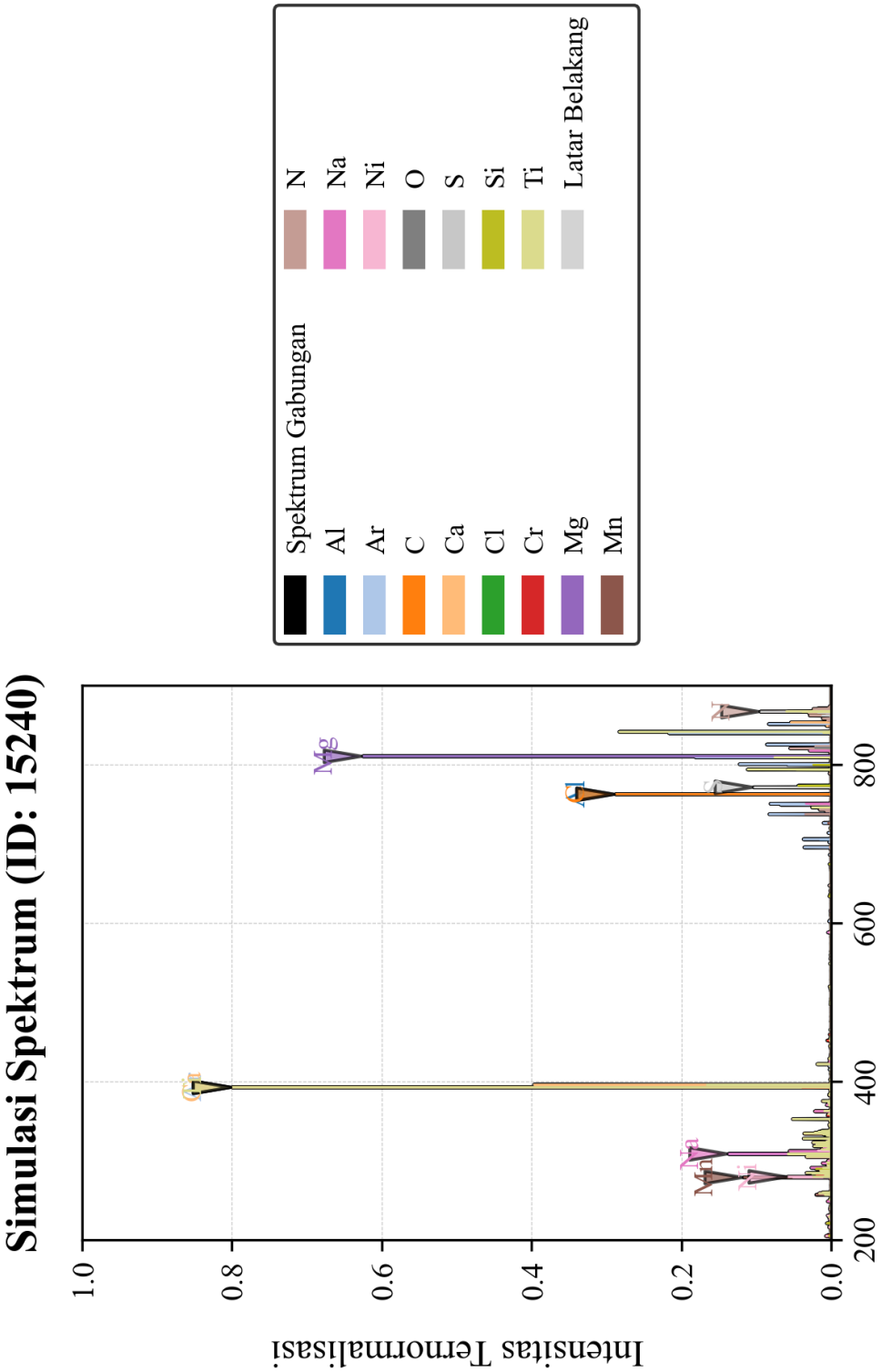

Figure 2: Spectral simulation plot for sample 15240 in full horizontal layout.

4 Sample: 14064

4.1 Simulation Parameters

Parameter	Value
Temperature (T)	6111 K
Electron Density (n_e)	1.12e+15 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	47.65

Table 5: Simulation parameters and additional metrics for sample 14064.

4.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	12.98	157.0
Ar	0.00	0.0
C	0.00	0.0
Ca	22.75	150.0
Cl	2.85	12.0
Co	0.00	0.0
Cr	3.80	397.0
Fe	0.00	0.0
Mg	13.14	183.0
Mn	11.93	681.0
N	0.00	0.0
Na	10.30	325.0
Ni	3.77	478.0
O	17.07	27.0
S	0.48	25.0
Si	0.93	177.0
Ti	0.00	0.0
background	N/A	2278.0
Total Positions		4890.0

Table 6: Composition analysis for sample 14064.

4.3 Spectral Plot

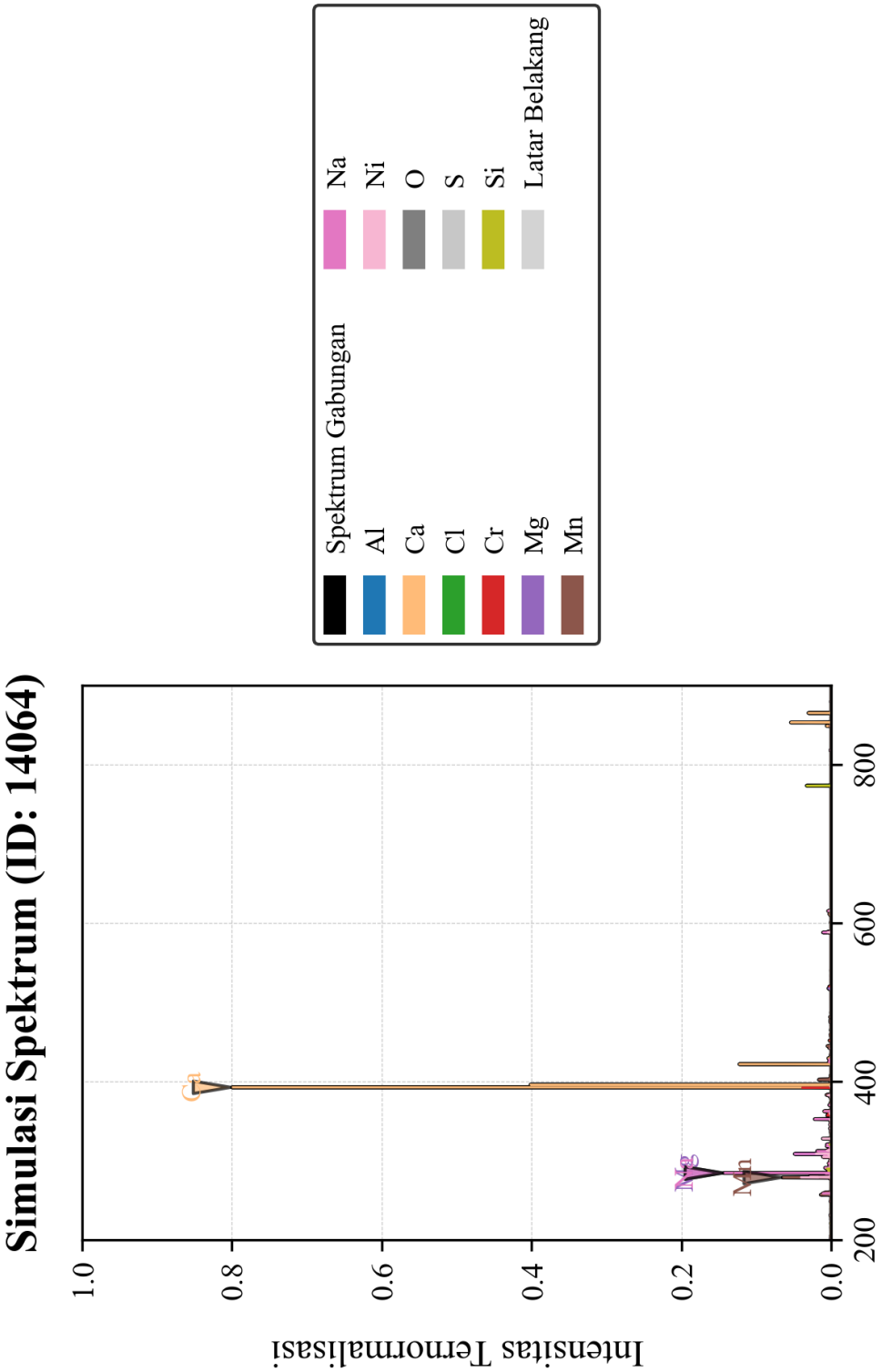

Figure 3: Spectral simulation plot for sample 14064 in full horizontal layout.

5 Sample: 8780

5.1 Simulation Parameters

Parameter	Value
Temperature (T)	9040 K
Electron Density (n_e)	1.10e+14 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	26.98

Table 7: Simulation parameters and additional metrics for sample 8780.

5.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	0.00	0.0
Ar	8.49	727.0
C	9.50	255.0
Ca	5.25	150.0
Cl	7.49	35.0
Co	9.50	1404.0
Cr	6.45	397.0
Fe	9.51	2740.0
Mg	6.46	341.0
Mn	5.52	706.0
N	8.48	401.0
Na	8.67	321.0
Ni	3.51	473.0
O	1.13	107.0
S	2.41	205.0
Si	7.65	223.0
Ti	0.00	0.0
background	N/A	511.0
Total Positions		8996.0

Table 8: Composition analysis for sample 8780.

5.3 Spectral Plot

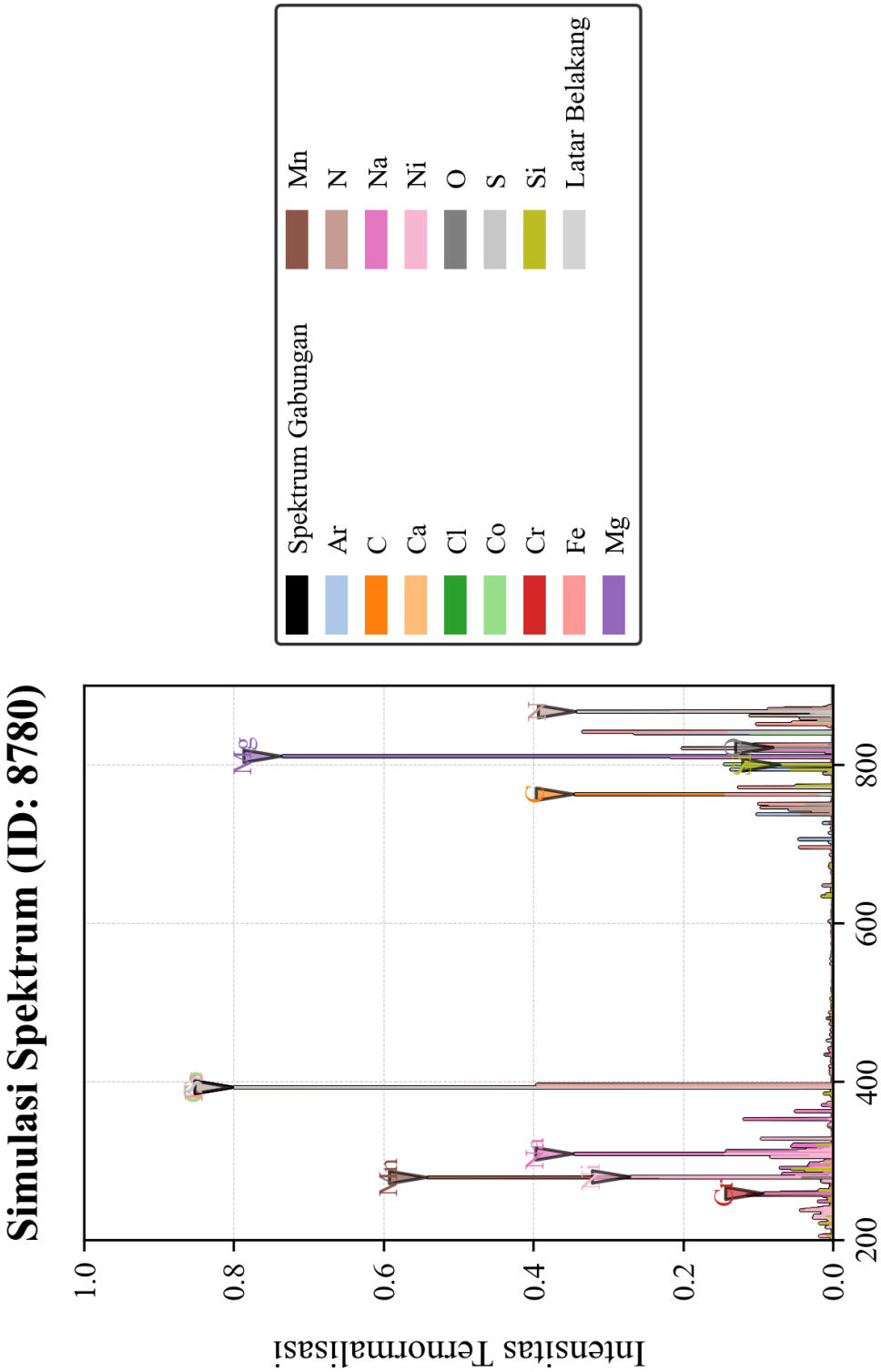

Figure 4: Spectral simulation plot for sample 8780 in full horizontal layout.

6 Sample: 10667

6.1 Simulation Parameters

Parameter	Value
Temperature (T)	7727 K
Electron Density (n_e)	1.79e+15 $\mathrm{cm}^{-}3$
Total Pixels	4096
Signal-to-Noise Ratio (SNR)	40.14

Table 9: Simulation parameters and additional metrics for sample 10667.

6.2 Composition Analysis

Element	Initial Composition (%)	Actual Positions
Al	6.22	279.0
Ar	0.00	0.0
C	0.00	0.0
Ca	8.18	150.0
Cl	5.11	12.0
Co	10.87	1404.0
Cr	3.96	397.0
Fe	2.63	2704.0
Mg	7.52	326.0
Mn	8.90	706.0
N	6.53	396.0
Na	4.75	322.0
Ni	8.70	478.0
O	9.24	101.0
S	5.65	61.0
Si	5.76	220.0
Ti	5.97	1184.0
background	N/A	590.0
Total Positions		9330.0

Table 10: Composition analysis for sample 10667.

6.3 Spectral Plot

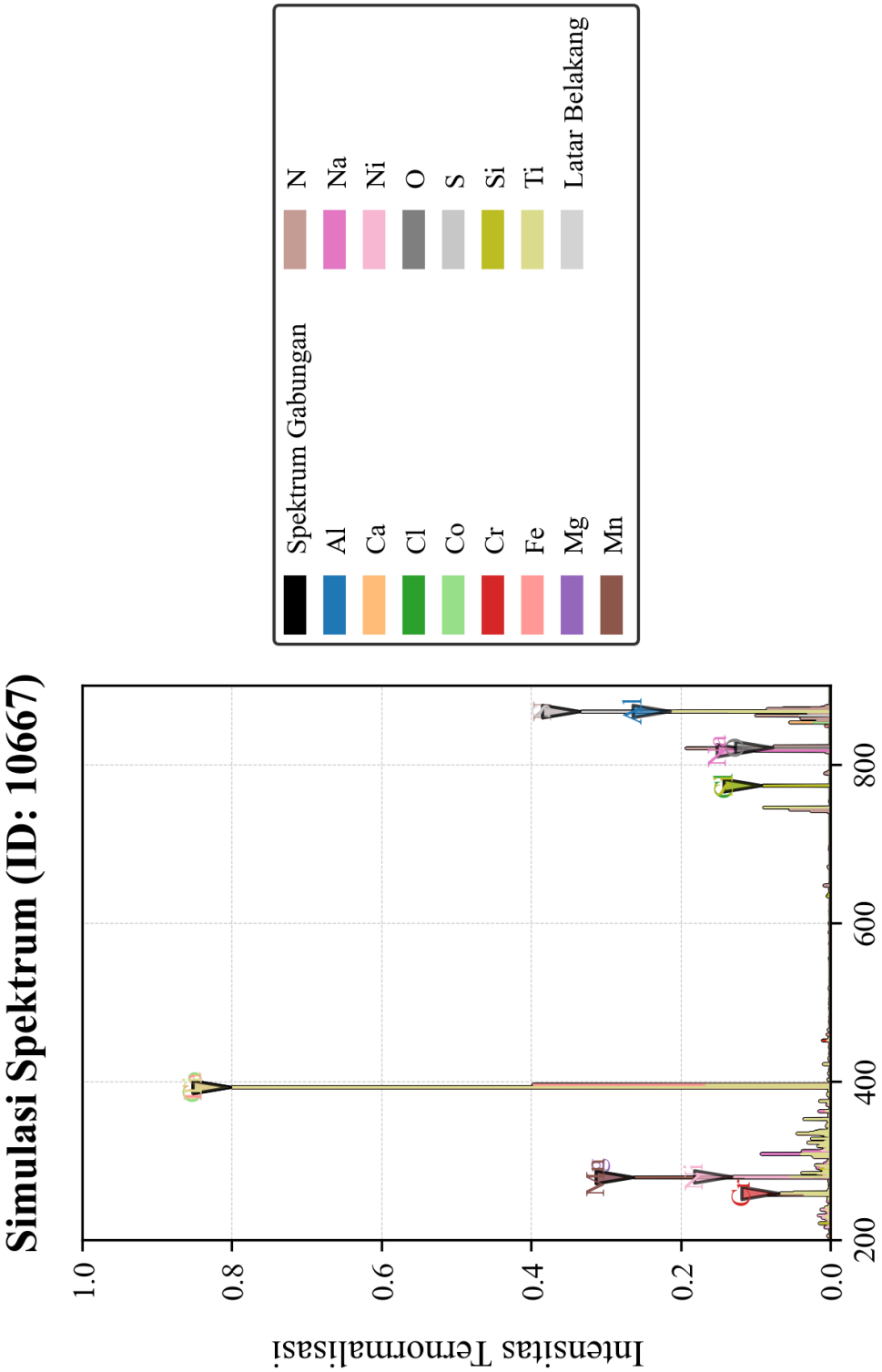

Figure 5: Spectral simulation plot for sample 10667 in full horizontal layout.